

Workshop 7: NoSQL databases

2026-04-22

Contents

1 Be positive! (5 min.)	1
2 Pokemon (30 min.)	1
3 Nobel prize (30 min.)	2
4 If you are done... ..	2

In this workshop, you will work in a team on a table with many other students around discovering and using NoSQL databases. It is a collective discussion to find a solution to the given problem, step by step by exchange and comments. Please use the site "mongodb playground" online, <https://mongoplayground.net/>. Please be positive, let all the people speak around the table, do not speak too loud, accept the comments for others, even negative comments can be treated as a way to improve your solution (always be positive). You can ask questions to me at any moment during the session. Välkomna, welcome, bienvenue...

1 Be positive! (5 min.)

Because, perhaps, you don't know each other around the table, take one minute per person to present yourself, explain who you are, where you come from, why you are taking this course/program and what you want to do in your future professional life. If everyone around the table is speaking Swedish, you can do it in Swedish, else, please go back to English.

2 Pokemon (30 min.)

- Please download the "workshop7-pokemon.json" file from Canvas. Copy and paste the JSON content to mongodb playground.
- Discuss with your group and answer the different questions, keep the code in a shared text file, and run your code into mongodb playground:
 1. How many pokemons are in the pokedex?
 2. Who are the pokemons having a "type" that contains "Fire"?
 3. Count how many pokemons have a "type" that contains "Fire"?
 4. Who are the pokemons having the types "Fire" and "Flying"?
 5. How many pokemons have a "type" that contains "Water" or "Electric"?
 6. Print the name only of the pokemon weighting "7.5 kg".
 7. Print the name of the evolution of Oddish.
 8. Print the number of pokemons having weight less than "1.0 kg".
 9. Print only the name and spawn_chance of the three first pokemons having the maximum spawn_chance.
 10. Print only the name and spawn_chance of the first three pokemons having the minimum spawn_chance.

3 Nobel prize (30 min.)

- Please download the "workshop7-nobel-prize-100.json" file from Canvas. Copy and paste the JSON content to mongodb playground.
1. Print all the persons having the bornCountry or the diedCountry equal to "France".
 2. Print the number (and only the number) of persons having the bornCountry or the diedCountry equal to "Sweden".
 3. What is the date of birth of Marie Curie?
 4. Can you print information on Marie Curie: firstname, lastname, bornCountry, diedCountry, date of birth?
 5. Can you print the shared (> 1) prize with the name, year and category of the prize?
 6. How many shared prizes in the collection?
 7. How many female in the collection?
 8. Print the firstname, surname, prize category and year of the persons who won at least two prizes?

4 If you are done...

In case you are done with the workshop very fast, you can add two other tables and add a REST API to play with the JOIN of the tables. :) B+